To interface an R307/R305 optical fingerprint sensor **and a** 1-channel relay module **with a** Raspberry Pi 4 **and simulate a fingerprint-authenticated door-lock system.**

Installation Manual

Hardware Requirements

1. Raspberry Pi Model A/B/B+
2. Jumper wires (Female to Male)
3. Relay Module
4. R307/R305 fingerprint sensor

- **Software Requirements**
  1. Raspbian Stretch OS

1. **Connect with Raspberry Pi's GPIO Pins.**
   - **Relay Module**

Device / Pin	Raspberry Pi 4	Physical Pin
Relay VCC	5V	2
Relay GND	GND	9
Relay IN	GPIO17	11

- **R307/R305 Fingerprint**

VCC	5V Pin 2	VCC
GND	GND Pin 6	GND
TX	GPIO15 / RXD, Pin 10	TX
RX	GPIO14 / TXD, Pin 8	RX

Software Installation
Step 1: Update Raspberry Pi packages
sudo apt update
sudo apt upgrade –y

Step 2: Install Python and serial support
sudo apt install -y python3 python3-pip python3-serial

Step 3: Install fingerprint library
pip3 install adafruit-circuitpython-fingerprint

Step 4: Enable Raspberry Pi UART
sudo raspi-config
Go to Interface Options ------ Serial Port
Login shell over serial: No
Serial port hardware: Yes
sudo reboot
After reboot, check the serial device:
ls -l /dev/serial0

If Raspberry Pi OS restricts system-wide pip installation
python3 -m venv fingerprint-env
source fingerprint-env/bin/activate
pip install pyserial adafruit-circuitpython-fingerprint RPi.GPIO

CODE:-

- ## Fingerprint Enrollment

```
import serial
import time
import adafruit_fingerprint

uart = serial.Serial("/dev/serial0", baudrate=57600, timeout=1)
finger = adafruit_fingerprint.Adafruit_Fingerprint(uart)

finger_id = int(input("Enter fingerprint ID (1-127): "))

print("Place finger on sensor...")
while finger.get_image() != adafruit_fingerprint.OK:
   time.sleep(0.1)

print("Image captured")

if finger.image_2_tz(1) != adafruit_fingerprint.OK:
   print("Image conversion failed")
   exit()

print("Remove finger")
time.sleep(2)

print("Place the same finger again...")
while finger.get_image() != adafruit_fingerprint.OK:
   time.sleep(0.1)

if finger.image_2_tz(2) != adafruit_fingerprint.OK:
   print("Image conversion failed")
   exit()

if finger.create_model() != adafruit_fingerprint.OK:
   print("Fingerprints did not match")
   exit()

if finger.store_model(finger_id) == adafruit_fingerprint.OK:
   print("Fingerprint enrolled successfully")
   print("Stored ID:", finger_id)
else:
   print("Failed to store fingerprint")
```

- **Main Door Lock Verification Program**

```
import serial
import time
import RPi.GPIO as GPIO
import adafruit_fingerprint

# Change this GPIO if required
RELAY_PIN = 17

# Active-LOW relay
RELAY_ON = GPIO.LOW
RELAY_OFF = GPIO.HIGH

GPIO.setmode(GPIO.BCM)
GPIO.setup(RELAY_PIN, GPIO.OUT, initial=RELAY_OFF)

try:
    uart = serial.Serial("/dev/serial0", baudrate=57600, timeout=1)
    finger = adafruit_fingerprint.Adafruit_Fingerprint(uart)

    print("Fingerprint Door Lock System")
    print("Place Finger")

    while True:
        result = finger.get_image()

        if result == adafruit_fingerprint.OK:
            print("Processing...")

            if finger.image_2_tz(1) != adafruit_fingerprint.OK:
                print("Fingerprint processing failed")
                time.sleep(1)
                continue

            if finger.finger_search() == adafruit_fingerprint.OK:
                print("Access Granted")
                GPIO.output(RELAY_PIN, RELAY_ON)

                print("Door Unlocked")
                time.sleep(5)

                GPIO.output(RELAY_PIN, RELAY_OFF)
                print("Door Locked")

            else:
                print("Access Denied")
                GPIO.output(RELAY_PIN, RELAY_OFF)

            time.sleep(1)
            print("Place Finger")

        elif result != adafruit_fingerprint.NO_FINGER:
            print("Fingerprint sensor error")
```

```
        time.sleep(1)

except Exception as e:
    print("Communication error:", e)

finally:
    GPIO.output(RELAY_PIN, RELAY_OFF)
    GPIO.cleanup()
```

Run the Main Program

python3 enroll.py

python3 door_lock.py